

Computational Fluid Dynamics

1. GENERAL

1.1 TITLE::Computational Fluid Dynamics

1.2 COURSE NUMBER (if known)::DE.CSM521.16

1.3 CREDITS:: 11 (3-0-2)

1.4 SEMESTER-OFFERED:: ODD

1.5 PRE-REQUISITES:: Fluid Dynamics, Partial Differential Equation

1.6 Course Committee Member:

2. OBJECTIVE::

The course aims to provide an understanding of basic concepts and introduces the fundamentals of finite difference methods and the notion of stability, accuracy and convergence of the methods. It is also aimed to develop an ability to use the techniques, skills, & engineering tools necessary for engineering practice by applying numerical methods to a "real-world" fluid-flow problem, integrating various numerical techniques in formulating a numerical solution method for that problem.

3. COURSE TOPICS::

UNIT I: Partial Differential Equations

Physical and Mathematical Classification, The Well-Posed Problem, System of Equations.

(03 Lectures)

UNIT II: Discretization Methods

Finite differences, Difference Representation of Partial differential Equations, Errors, Consistency, Stability, Convergence for Marching Problems, Conservation Form, Fourier or von Neumann Stability Analysis.

(06 Lectures)

UNIT III: Methods for Hyperbolic Partial Differential Equations

Euler Explicit and Implicit Methods, Upstream Differencing Method, Lax Method, Leap-Frog Method, Lax-Wendroff Method, McCormack Method, Rusanov Method.

(06 Lectures)

UNIT IV: Methods for Parabolic Partial Differential Equations

Simple Explicit and Implicit Method, Richardson's Method, Crank-Nikolson Method, ADI and ADE Methods.

(05 Lectures)

UNIT V: Methods for Elliptic Partial Differential Equations

Finite-Difference Representations of Laplace's Equation, Direct and Iterative Methods for Solving System of Linear Algebraic Equations.

(05 Lectures)

UNIT VI: Governing Equations of Fluid Mechanics and Heat Transfer

Continuity Equation, Momentum Equation, Energy Equation, Equation of State, Boundary Layer Approximation for Steady Incompressible Flow, Euler Equations, Shock Equations.(05 Lectures)

UNIT VII: Numerical Methods for Inviscid Flow Equations

Method of Characteristics for Linear and Nonlinear System of Equations, Shock Capturing Methods.

(05 Lectures)

UNIT VIII: Numerical Methods for Boundary Layer Type Equations

Explicit and Implicit Methods, Crank Nicolson and DuFort-Frankel Methods. (04 Lectures)

4.READINGS

4.1 TEXTBOOK::

1. Dale A. Anderson, John C. Tannehill and Richard H. Platcher.. Computational Fluid Mechanics and Heat Transfer; McGraw Hill Book Company.

2. C.A.J. Fletcher. Computational Techniques for Fluid dynamics: Vol – I & II, Springer-Verlag, Berlin.

4.2 *REFERENCE BOOKS::

W.F.Ames.. Numerical Method for Partial Differential Equation, Academic Press.

5. OTHER SESSIONS

- 5.1 *TUTORIALS:: Nil
 5.2 *LABORATORY:: Yes (2 hrs.)
 5.3 *PROJECT:: To demonstrate the use of methods in real life engineering problems.

6. ASSESSMENT (indicative only)

- 6.1 HA:: 0%
 6.2 QUIZZES-HA:: 0%
 6.3 PERIODICAL EXAMS:: 40%
 6.4 *PROJECT:: 20%
 6.5 FINAL EXAM:: 40%

7. OUTCOME OF THE COURSE::

At the end of the course the students should be able to

- apply knowledge of math and science to engineering by describing a continuous fluid-flow phenomena in a discrete numerical sense.
- use the techniques, skills, & engineering tools necessary for engineering practice by applying numerical methods to a "real-world" fluid-flow problem, integrating various numerical techniques in formulating a numerical solution method for that problem.
- analyze and interpret data obtained from the numerical solution of fluid flow problems.
- communicate effectively by writing the term project in a structured technical report format.

8. *EXPECTED ENROLLMENT FOR THE COURSE::40

9. *DEPARTMENTS OTHER THAN YOUR OWN TO WHICH THIS COURSE WOULD BE OF INTEREST::Civil Engineering, Mechanical Engineering, Chemical Engineering.

10. *ANY OTHER REMARKS::